

4G Module Introduction

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I. SIM7600G-H-M2 Module Overview

SIM7600G-H-M2 is an LTE Cat.4 industrial-grade 4G universal M.2 module launched by SIMCom. It supports global 4G/3G/2G network communication, with theoretical peak download speed of 150Mbps and upload speed of 50Mbps. It adopts standard M.2 B-KEY form factor, integrates easy-to-use AT commands, and is compatible with mainstream systems such as Windows/Linux/Android. It is widely used in IoT, industrial routers, vehicles, security, embedded devices and other scenarios.

II. SIM7600G-H Chip Specifications

General Features

Frequency Bands	LTE-TDD B34/B38/B39/B40/B41
	LTE-FDD B1/B2/B3/B4/B5/B7/B8/B12/B13/B18/B19/B20/B25/B26/B28/B66
	UMTS/HSPA+ B1/B2/B4/B5/B6/B8/B19
	GSM/GPRS/EDGE 850/900/1800/1900 MHz
Supply Voltage	3.4V ~ 4.2V, Typ: 3.8V
Control Via AT Commands	
Operation temperature	-40°C ~ +85°C
Dimensions	30.0*30.0*2.9mm
Weight	5.7±0.1g

Data Transfer

LTE Cat4	Uplink up to 50Mbps,
	Downlink up to 150Mbps
HSPA+	Uplink up to 5.76Mbps
	Downlink up to 42 Mbps
UMTS	Uplink/Downlink up to 384Kbps
EDGE	Uplink/Downlink up to 236.8Kbps
GPRS	Uplink/Downlink up to 85.6Kbps

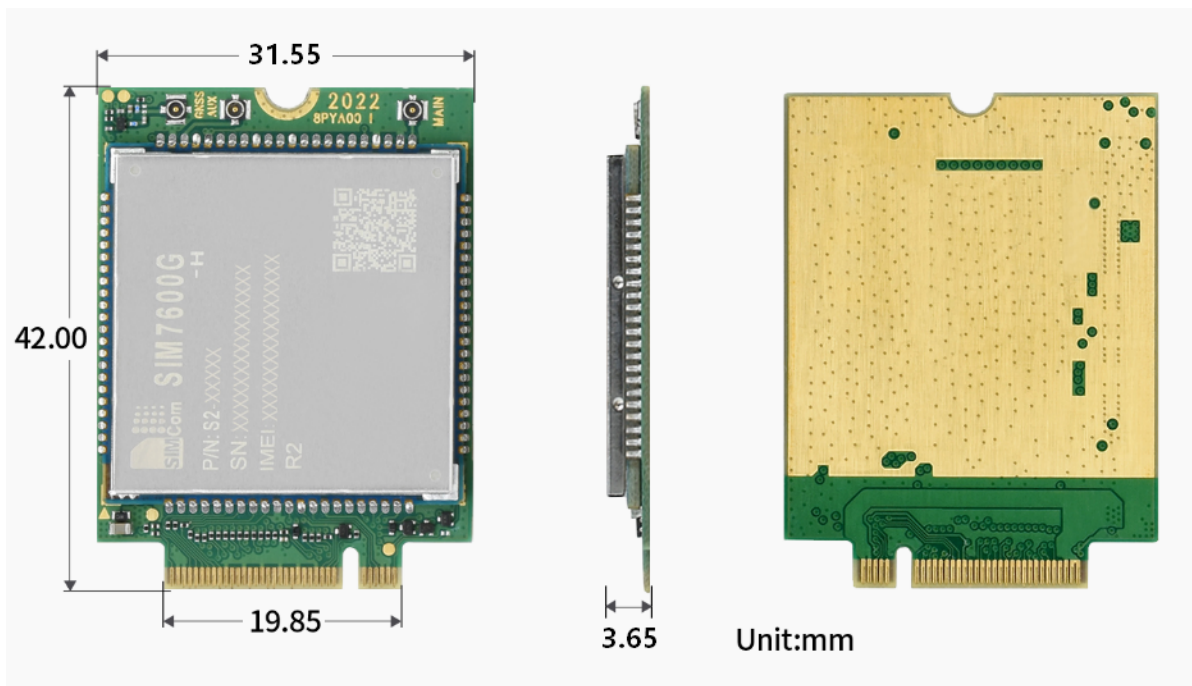
III. SIM7600G-H-M2 Pin Functions

Note: The SIM7600G-H-M2 module uses M.2 B KEY interface, which is not compatible with MINI-PCIE interface. Please distinguish carefully.

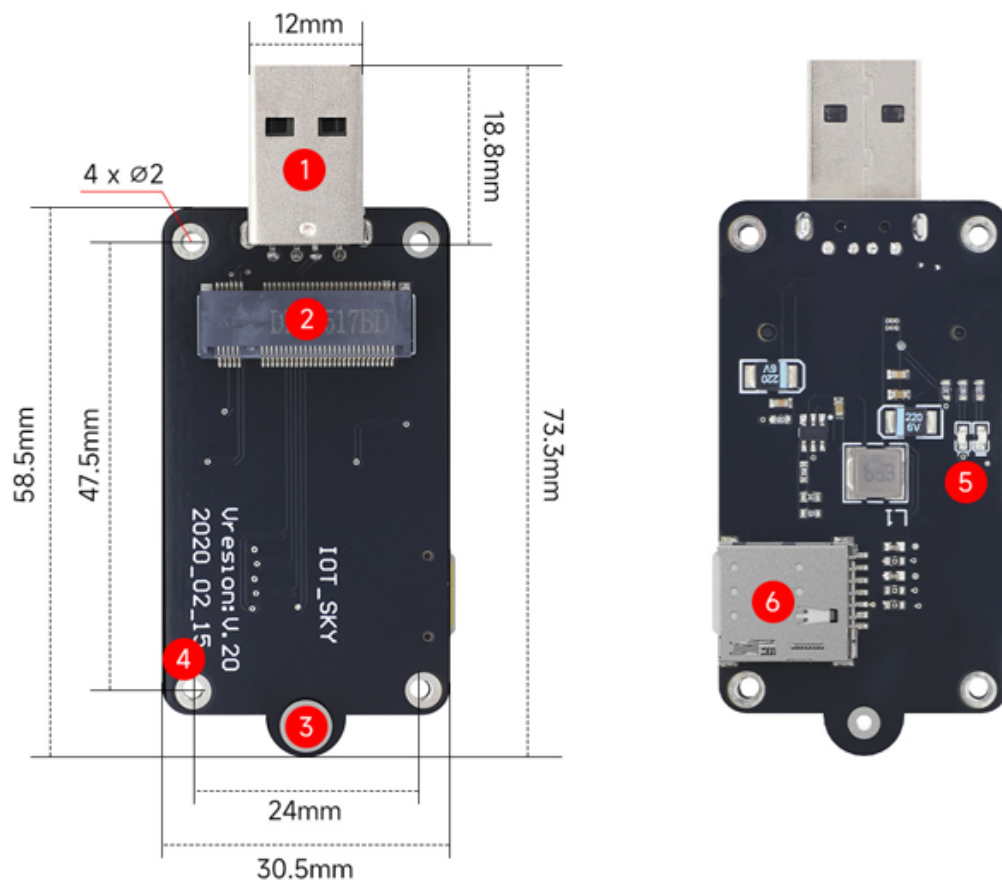


74	VBAT	CONFIG_2=GND	75
72	VBAT	GND	73
70	VBAT	GND	71
68	NC	CONFIG_1=GND	69
66	USIM_DET(I)(1.8V)	RESET#(I)(1.8V)	67
64	GPIO3(IO)(1.8V)	SPI_CLK	65
62	GPIO77(IO)(1.8V)	SPI_MISO	63
60	UART_TXD(O)(1.8V)	SPI_MOSI	61
58	UART_RXD(I)(1.8V)	SPI_CS	59
56	UART_CTS(O)(1.8V)	GND	57
54	UART_RTS(I)(1.8V)	NC	55
52	UART_DTR(I)(1.8V)	NC	53
50	GPIO40(IO)(1.8V)	GND	51
48	GPIO41(IO)(1.8V)	NC	49
46	GPIO43(IO)(1.8V)	NC	47
44	GPIO44(IO)(1.8V)	GND	45
42	I2C_SDA(IO/OD)(1.8V)	NC	43
40	I2C_SCL(O/OD)(1.8V)	NC	41
38	NC	GND	39
36	USIM_VDD	NC	37
34	USIM_DATA	NC	35
32	USIM_CLK	GND	33
30	USIM_RST	NC	31
28	PCM_CLK(O)(1.8V)	NC	29
26	W_DISABLE2_N(I)(3.3V)	GND	27
24	PCM_OUT(O)(1.8V)	DPR(I)(1.8V)	25
22	PCM_IN(I)(1.8V)	WoWWAN(OD)(1.8V/3.3V)	23
20	PCM_SYNC(O)(1.8V)	CONFIG_0=GND	21
	Notch	Notch	
	Notch	Notch	
	Notch	Notch	
	Notch	Notch	
10	LED1#(OD)(3.3V)	GND	11
8	W_DISABLE1_N(I)(3.3V)	USB_D-	9
6	FUL_CARD_POWER_OFF#(I)(1.8/3.3V)	USB_D+	7
4	VBAT	GND	5
2	VBAT	GND	3
		CONFIG_3=NC	1

IV. SIM7600G-H-M2 Dimensions



V. USB Expansion Board Specifications

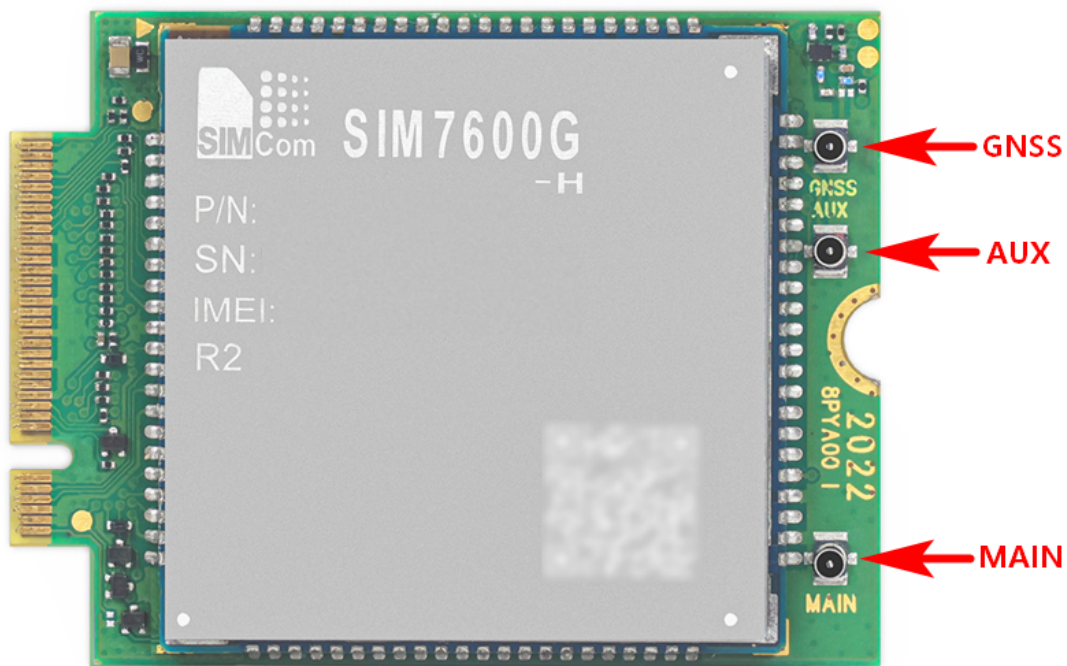


No.	Interface Name	Function
①	USB 2.0 Interface	Used to connect to motherboard USB port for communication
②	M.2 B-KEY Interface	Used to connect SIM7600G-H-M2 module
③	Module Fixing Screws and Patch Nuts	Fix the module
④	Four Mounting Holes	Used to install metal bracket
⑤	Indicator Lights	Indicate power and signal connection status
⑥	Nano-SIM Socket	Insert Nano-SIM card

VI. Antenna Specifications

The antenna interface model of SIM7600G-H-M2 module is IPEX 4th generation interface.

Antenna Interface	Function	Remarks
MAIN	Main antenna, must be connected	LTE/4G, 3G, 2G transmit + receive main path, responsible for core services such as network registration, data transmission/reception, calls, etc. Must be connected, otherwise the module cannot access the network or register with base stations
AUX	Diversity antenna, recommended	LTE/4G receive diversity (receive only, no transmit), forms diversity reception with MAIN to improve sensitivity, speed and stability under weak signals. Spatial isolation from MAIN >5cm.
GNSS	Positioning antenna, connect as needed	Dedicated to satellite positioning such as GPS, Beidou, GLONASS, Galileo, providing positioning information such as latitude/longitude, speed, time. Not required if positioning is not needed.



Rubber Rod Antenna Specifications



Parameter Name	Data
Frequency Range (MHz)	700-2700
Bandwidth (MHz)	2000
Input Impedance (Ω)	50
V.S.W.R	≤ 3.0
Gain (dBi)	4
Max Input Power (W)	10
Antenna Length (mm)	195
Connect Type	SMA female
Radome Color	Black
Weight (g)	22

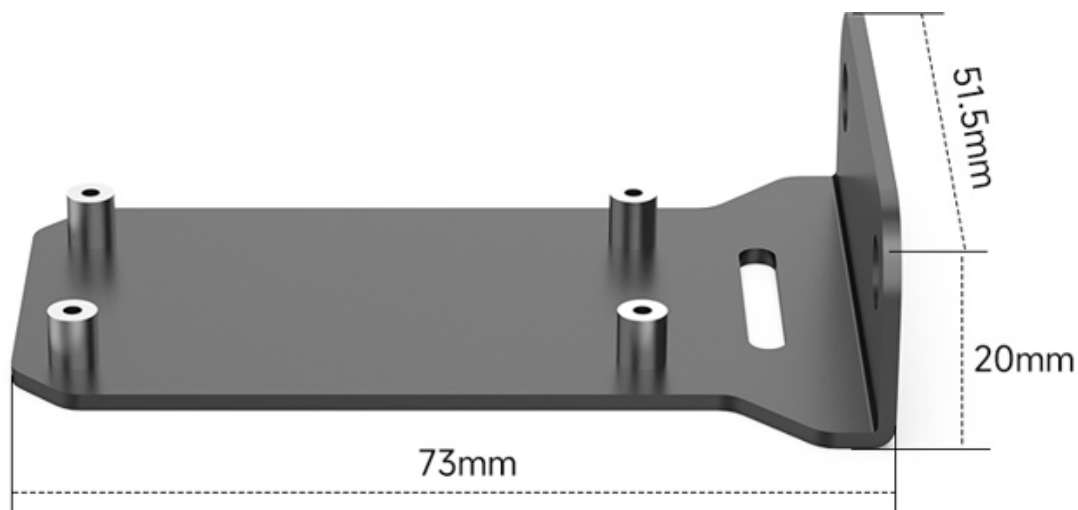
VII. Nano-SIM Card Selection

Supports 4G networks of three major domestic carriers (China Telecom, China Mobile, China Unicom).

For overseas operators, LTE-TDD, LTE-FDD, UMTS/HSPA+, GSM/GPRS/EDGE communication standards are required.

SIM card voltage support: 1.8V/3.3V

VIII. Aluminum Alloy Fixing Plate Specifications



Aluminum Alloy Fixing Plate	Parameters
Material	Aluminum alloy
Process	Aluminum alloy bending
Surface Treatment	Anodizing
Dimensions	7351.520mm
Weight	16g